PRODUCT REQUIREMENTS DOCUMENT (PRD)

Futsal Court Booking Platform

Module 4: Court Booking & Digital Payment

Document Information	Details
Product Name	Futsal Court Booking Platform
Module	Court Booking & Digital Payment
Document Version	1.0
Status	Draft
Product Role	Technical Lead Product Manager
Primary Target User	Dane, University Student & Futsal Team Captain
Platform	Web Application
Target Market	Indonesia

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1. Product Overview

1.1 Product Vision

To provide a convenient, reliable, and transparent digital platform that allows futsal enthusiasts to discover, compare, reserve, and pay for futsal courts online without relying on manual phone calls or physical visits.

The platform aims to simplify the entire booking journey, from finding an available court to completing payment and receiving a digital booking confirmation.

1.2 Problem Statement

Futsal players and team captains often experience difficulties securing court reservations, particularly during peak evening hours between 18:00 and 22:00.

Traditional booking processes rely heavily on phone calls, messaging applications, or direct visits to venues. These methods create several problems:

- Limited visibility into real-time court availability.
- Risk of double bookings caused by manual reservation management.
- Slow booking confirmation and payment verification.
- Lack of transparent information about court prices, surfaces, facilities, and quality.
- Difficulty collecting and coordinating payments among team members.

These challenges create unnecessary friction for team captains who need to organize regular futsal matches.

1.3 Proposed Solution

Develop a web-based futsal booking platform with real-time availability, secure digital payments, and team bill-splitting capabilities.

The platform will allow users to search and compare courts, select available time slots, temporarily reserve a slot during checkout, complete payments, and receive an instant digital booking ticket.

Captains can also distribute booking costs among team members through a Split Bill feature, reducing the burden of collecting payments individually.

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2. Target Persona

2.1 Primary Persona: Dane

Name: Dane
Age: 20
Occupation: University Student
Role: Futsal Enthusiast and Amateur Team Captain

2.2 Persona Profile

Dane regularly organizes futsal matches with university friends and amateur teams. He is responsible for finding suitable venues, coordinating match schedules, securing reservations, and collecting payments from teammates.

As a digitally active student, Dane prefers mobile-friendly online services that provide fast information, transparent pricing, and reliable booking confirmations.

2.3 Goals

- Find available futsal courts near his preferred location.
- Compare prices, court surfaces, reviews, and amenities.
- Secure evening slots without contacting multiple venues.
- Divide booking costs fairly among team members.
- Receive immediate confirmation after successful payment.
- Manage upcoming and previous bookings in one place.

2.4 Pain Points

- Evening slots are frequently unavailable.
- Court availability information is often outdated.
- Manual booking can result in double reservations.
- Venue prices and facilities are not always clearly displayed.
- Collecting money from teammates takes time.
- Booking confirmations may be delayed or unreliable.

2.5 User Needs

Dane needs a centralized booking system that provides accurate availability, transparent venue information, secure payments, and a straightforward way to coordinate team expenses.

2.6 Persona Success Criteria

Dane considers the platform successful when he can find and book a suitable court within a few minutes, complete payment with minimal friction, and share the cost with his teammates without excessive follow-up.

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3. Product Goals and Objectives

3.1 Primary Goals

1. Simplify the process of discovering and booking futsal courts.
2. Prevent double bookings through real-time availability and slot locking.
3. Reduce manual payment verification through payment gateway integration.
4. Simplify team payment coordination using Split Bill.
5. Improve trust through transparent venue information and digital booking confirmations.

3.2 Product Objectives

- Make court search and slot selection fast and intuitive.
- Provide a reliable reservation lifecycle from checkout to confirmation.
- Support commonly used Indonesian digital payment methods.
- Give captains clear visibility into booking and team payment status.
- Increase customer retention through a smooth repeat-booking experience.

3.3 Product Scope

The MVP focuses on the core transaction journey: **Search → Select → Reserve → Pay → Confirm → Check In.**

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4. User Stories

4.1 Primary User Story

As a team captain trying to book a court for a weekly match, I want to check live court availability and book instantly while allowing team members to pay their respective shares, so that I can secure the best time slot without stress and ensure all payments are collected.

4.2 Supporting User Stories

Court Discovery

- As a user, I want to search for courts by location, date, time, and price so that I can find a suitable venue.
- As a user, I want to compare court surfaces, amenities, and reviews so that I can make an informed decision.

Reservation

- As a user, I want to view real-time availability on a calendar so that I can choose an available slot.
- As a user, I want my selected slot to be temporarily locked during checkout so that another user cannot reserve it simultaneously.

Payment

- As a user, I want to pay using QRIS or bank transfer so that I can complete my booking conveniently.
- As a captain, I want to split the total cost among team members so that each person can contribute fairly.
- As a captain, I want to monitor payment progress so that I know whether the booking is fully funded.

Confirmation and Management

- As a user, I want to receive a digital ticket and QR code after successful payment so that I can check in at the venue.
- As a user, I want to view my booking history so that I can manage upcoming and previous reservations.

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5. Functional Requirements

FR-01: User Registration and Authentication

Description: The system must allow users to create accounts and authenticate securely.

Requirements:

- Support registration, login, and logout.
- Support Captain and Regular Member account roles.
- Allow users to manage their basic profile information.
- Restrict access to booking and payment information according to account permissions.

Acceptance Criteria:

- Users can register and log in with valid credentials.

- Invalid credentials produce a clear error message.
- Users can access only the information and actions permitted by their roles.

FR-02: Court Search and Filtering

Description: The system must provide search and filtering tools for finding suitable futsal courts.

Requirements:

- Search by location, date, and preferred time.
- Filter by price range, court type, surface, and amenities.
- Support surface types such as vinyl and parquet.
- Display venue name, location, hourly price, available slots, facilities, and user reviews where available.
- Show an appropriate empty state when no courts match the selected criteria.

Acceptance Criteria:

- Users can apply multiple filters simultaneously.
- Search results reflect the selected criteria.
- Each result provides sufficient information to compare venues.
- Search results load within two seconds under defined normal operating conditions.

FR-03: Real-Time Availability and Slot Reservation

Description: The system must allow users to view and reserve available one-hour court slots.

Requirements:

- Display court availability through a calendar interface.
- Allow users to select a date and available one-hour slot.
- Temporarily lock a selected slot for five minutes during checkout.
- Prevent overlapping confirmed reservations for the same court and time.
- Automatically release expired reservations.
- Display slot status, such as Available, Temporarily Held, Booked, or Unavailable.

Acceptance Criteria:

- Unavailable slots cannot be selected.
- Concurrent booking attempts cannot produce two confirmed reservations for the same slot.
- A slot is released when its checkout hold expires without successful payment.
- The system revalidates slot availability before confirming a reservation.

FR-04: Checkout and Payment Processing

Description: The system must enable users to pay for reservations using supported digital payment methods.

Requirements:

- Display a booking summary before payment.
- Show the selected venue, date, time, duration, and total price.
- Support QRIS and virtual account (VA) bank transfers through a payment gateway.
- Update payment status automatically using verified gateway notifications.
- Maintain payment transaction records.
- Display pending, successful, failed, expired, and refunded statuses where applicable.
- Handle duplicate payment notifications safely.

Acceptance Criteria:

- Users can review the total amount before initiating payment.
- A verified successful payment updates the transaction status.
- Payment failure or expiry generates an appropriate message.
- The system does not confirm a booking solely because a user returns to a success page.
- Duplicate gateway notifications do not create duplicate bookings or payment records.

FR-05: Split Bill

Description: The system must allow captains to distribute booking costs among team members.

Requirements:

- Allow captains to add or select team members for a booking.
- Calculate equal payment shares by default.
- Display the total cost, individual share, and outstanding balance.
- Generate individual payment requests or payment links for team members.
- Track each member's payment status.
- Notify the captain when a member completes payment.
- Allow the captain to cover unpaid shares.

Acceptance Criteria:

- The sum of allocated shares equals the booking total.
- Each team member can view their assigned payment amount and status.
- Successful payments update the outstanding balance.
- The captain can identify unpaid team members.
- Any rounding difference is allocated transparently so the total remains exact.

Business Rule: Split Bill payment deadlines and cancellation consequences must follow the venue's configured policy. The captain must be informed of the outstanding balance and any deadline before booking confirmation.

FR-06: Booking Management and History

Description: The system must provide a dashboard for managing reservations.

Requirements:

- Display upcoming, completed, cancelled, and expired bookings.
- Display reservation details and payment status.
- Allow users to view booking information and digital tickets.
- Provide cancellation or refund requests according to venue policy.
- Allow captains to view Split Bill progress for their reservations.

Acceptance Criteria:

- Booking history displays the correct reservation and payment statuses.
- Users can open a booking to view its details.
- Cancellation actions respect the applicable cancellation policy.
- Changes in booking status are reflected in the dashboard.

FR-07: Digital Ticket and Venue Check-In

Description: The system must generate a digital booking ticket after a reservation is confirmed.

Requirements:

- Generate a unique booking reference.
- Generate a QR code associated with the booking.
- Display the venue, date, start time, duration, and booking reference.
- Provide a downloadable or shareable ticket.
- Provide an option to add the booking to a calendar using a standard calendar invitation or compatible calendar file.
- Allow venue staff or an authorized check-in interface to validate tickets.

Acceptance Criteria:

- Confirmed bookings have a unique ticket.
- Tickets cannot be generated as valid confirmed tickets for failed or expired bookings.
- The QR code can be validated against the booking record.
- Used, cancelled, or invalid tickets are handled appropriately during check-in.

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6. User Flow

6.1 Primary Booking Flow

1. User logs in to the platform.
2. User searches for a futsal court by location, date, and time.
3. User applies filters and compares venue information.
4. User selects a court and an available one-hour slot.
5. System temporarily locks the slot for five minutes.
6. User reviews the booking summary.

7. User chooses individual payment or Split Bill.
8. User completes payment, or team members pay their allocated shares according to the selected payment policy.
9. System verifies payment and updates the reservation status.
10. System generates a digital ticket with a QR code.
11. User receives booking confirmation and a calendar invitation.
12. User presents the QR code at the venue for check-in.

6.2 Split Bill Flow

1. Captain selects Split Bill during checkout.
2. Captain adds team members.
3. System calculates the allocated payment shares.
4. System generates payment requests.
5. Members complete their assigned payments.
6. System updates the payment status for each member.
7. Captain views the remaining balance.
8. The booking follows the configured confirmation and settlement policy.
9. Once the reservation is confirmed, the digital ticket becomes available.

Important: The product must define whether the reservation is confirmed after the captain pays the full amount, after all team members pay, or after the captain guarantees the outstanding balance. This decision must be consistent with venue policy.

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7. Business Rules

BR-01: Slot Exclusivity

A court cannot have more than one active confirmed booking for overlapping time periods.

BR-02: Temporary Reservation

A selected slot is held for five minutes during checkout. The hold must expire automatically if payment is not completed within the permitted window.

BR-03: Payment Verification

Booking confirmation depends on verified payment status and a valid reservation state.

BR-04: Split Bill Allocation

The total of all assigned shares must equal the booking price, including any applicable service fees or taxes.

BR-05: Incomplete Split Bill

If team members have not completed their payments before the configured deadline, the captain must settle the outstanding amount or the booking may be cancelled according to venue policy.

BR-06: Cancellation and Refunds

Cancellation eligibility, refund amounts, and refund processing times must be displayed before users confirm their bookings.

BR-07: QR Ticket Validation

A booking QR code must reference a valid booking and must not expose sensitive personal or payment information.

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8. Edge Cases and Error Handling

EC-01: Simultaneous Booking Attempts

Scenario: Two users attempt to reserve the same court and time slot simultaneously.

Expected Behavior:

- The backend must atomically acquire the slot hold for only one user.
- The other user receives a message that the slot is no longer available.
- The user can return to the calendar and select another slot.
- The system must enforce uniqueness or equivalent transactional protection at the database level.

EC-02: Payment Timeout

Scenario: The user does not complete payment within five minutes.

Expected Behavior:

- The reservation hold expires.
- The slot returns to the available pool if no other valid reservation exists.
- The booking order is marked as expired.
- The user receives a clear expiry message.

Additional Safeguard: If a delayed payment notification arrives after the hold has expired, the system must reconcile the payment without confirming an already-reassigned slot. The user must receive an appropriate refund or resolution according to the payment and venue policies.

EC-03: Incomplete Split Bill

Scenario: Some team members fail to pay their shares before the deadline.

Expected Behavior:

- The captain receives an outstanding balance summary.
- The system sends reminders according to the notification policy.
- The captain can pay the remaining balance if permitted.
- The system applies the configured settlement or cancellation policy if the balance remains unpaid.

EC-04: Payment Gateway Failure

Scenario: The payment gateway becomes unavailable or returns an error.

Expected Behavior:

- The system displays a clear failure or pending status.
- Users can retry or select another supported payment method when appropriate.
- The system prevents duplicate transactions from repeated submissions.
- The slot remains held only for the permitted reservation window.

EC-05: Stale Availability Data

Scenario: A court becomes unavailable after search results are displayed.

Expected Behavior:

- The system rechecks availability when the user selects the slot.
- The user is informed if the slot can no longer be reserved.
- The calendar refreshes with the latest status.

EC-06: Duplicate Payment Notification

Scenario: A payment gateway sends the same notification more than once.

Expected Behavior:

- The notification is verified and processed idempotently.
- The transaction and booking are updated only as appropriate.
- No duplicate booking or ticket is created.

EC-07: Invalid or Reused QR Ticket

Scenario: A cancelled, invalid, or previously checked-in ticket is scanned.

Expected Behavior:

- The check-in interface displays an invalid, cancelled, or already-used status.
- Only authorized venue personnel can perform check-in validation.
- Check-in actions are logged for auditing.

9. Non-Functional Requirements (NFR)

NFR-01: Performance

- Search results and calendar availability should render within two seconds under normal defined load conditions.
- Booking operations should provide clear loading and processing feedback.
- The system must be tested for concurrent traffic during peak hours, particularly 18:00–22:00.
- Performance targets must specify the expected number of concurrent users and measurement percentile before release.

NFR-02: Security

- All network traffic must use HTTPS/TLS.
- Sensitive user data must be protected at rest and in transit.
- Authentication and role-based authorization must be enforced.
- Payment processing must use a compliant payment gateway.
- PCI DSS obligations must be addressed according to the platform's actual card-data handling and payment integration model.
- Payment notifications must be authenticated or cryptographically verified where supported.
- Sensitive information must not be exposed through booking QR codes or public URLs.

NFR-03: Reliability and Availability

- Target service availability: 99.9% monthly uptime.
- Automated database backups must be maintained.
- Recovery procedures must be documented and tested.
- Reservation and payment state changes must be transactionally consistent.
- Monitoring and alerting must cover booking failures, payment failures, and availability errors.

NFR-04: Scalability

- The platform must support increasing numbers of venues, users, bookings, and concurrent searches.
- Availability checks and reservation locking must remain reliable under peak load.
- Load testing must validate the target concurrency before production launch.

NFR-05: Usability and Accessibility

- The interface must be responsive across desktop, tablet, and mobile browsers.
- Booking statuses and errors must use clear, understandable language.
- Forms must provide validation and actionable error messages.
- Important information must not rely on color alone.

- The booking flow should minimize unnecessary steps.

NFR-06: Data Integrity

- A court and time slot must not have overlapping confirmed reservations.
- Payment and booking statuses must remain consistent.
- Audit logs must record critical reservation, payment, cancellation, and check-in changes.
- Backup restoration must preserve the integrity of booking and transaction records.

NFR-07: Privacy

- Collect only personal information necessary for the service.
- Provide an appropriate privacy notice and consent where required.
- Restrict access to personal and payment-related records.
- Establish data retention and deletion procedures in accordance with applicable Indonesian regulations.

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10. Success Metrics and KPIs

KPI-01: Booking Conversion Rate

Definition: Percentage of users who complete a booking after initiating a court search.

Formula:

Booking Conversion Rate = (Users completing a booking ÷ Users initiating a court search) × 100%

Objective: Establish a baseline during beta testing and improve conversion through better availability visibility and a simpler checkout experience.

KPI-02: Split Bill Adoption Rate

Definition: Percentage of eligible bookings that use the Split Bill feature.

Formula:

Split Bill Adoption Rate = (Eligible bookings using Split Bill ÷ Total eligible bookings) × 100%

Objective: Measure whether the feature solves the payment coordination problem for team captains.

KPI-03: Repeat Booking Rate

Definition: Percentage of team captains who make another booking within 30 days of their initial booking.

Formula:

Repeat Booking Rate = (Captains who book again within 30 days ÷ Captains eligible for the 30-day measurement window) × 100%

Objective: Evaluate user satisfaction, reliability, and long-term retention.

Supporting Metrics

- Search-to-slot-selection rate.
- Checkout abandonment rate.
- Successful payment rate.
- Average time to complete a booking.
- Slot conflict rate.
- Booking cancellation rate.
- Split Bill settlement rate.
- Payment failure and expiry rate.
- QR check-in success rate.

Measurement Requirement: Establish baseline values before setting numerical improvement targets. KPI calculations must use consistent event definitions and exclude users or bookings that are not eligible for the relevant measurement window.

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11. MVP Scope

11.1 In Scope

- User registration and authentication.
- Captain and Regular Member roles.
- Court search and filtering.
- Court details, pricing, amenities, and reviews where available.
- Real-time court availability.
- One-hour slot selection and temporary reservation locking.
- Checkout and booking summary.
- QRIS and virtual account bank transfer integration.
- Automated payment status updates.
- Split Bill creation, payment tracking, and balance monitoring.
- Booking history and management dashboard.
- Digital booking ticket and QR code generation.
- Calendar invitation.
- Basic cancellation, refund, and venue check-in workflows.
- Booking, payment, and reservation audit records.

11.2 Out of Scope

- Community sparring and random team matchmaking.
- Marketplace for futsal shoes, jerseys, balls, and other equipment.
- Advanced tournament management and bracket generation.
- Automated team formation.
- Advanced loyalty and membership programs.
- Predictive court recommendations and dynamic pricing.

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12. Dependencies and Assumptions

Dependencies

- A payment gateway supporting QRIS and bank virtual accounts.
- Reliable venue inventory and availability data.
- A backend reservation service with atomic slot locking.
- A notification service for payment requests and booking updates.
- QR code generation and authorized ticket validation.
- Calendar invitation or calendar-file generation.
- A venue management interface or integration to keep court availability accurate.

Assumptions

- Venues provide accurate court schedules, pricing, and operating hours.
- All bookable slots have a defined duration and pricing policy.
- The platform can receive payment status updates from the payment gateway.
- Team members can access payment requests through the platform or supported links.
- Venue owners establish cancellation, refund, and Split Bill settlement policies.
- The initial launch prioritizes the Indonesian market and its common digital payment methods.

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13. Release Plan

Phase 1: MVP Foundation

- User authentication and account roles.
- Venue listings, court details, and search filters.
- Availability calendar and reservation locking.
- Booking summary and payment integration.
- Booking confirmation and digital tickets.

Phase 2: Team Payment Coordination

- Split Bill creation and allocation.
- Member payment requests.
- Payment tracking and outstanding balance dashboard.

- Notifications and payment deadline enforcement.

Phase 3: Reliability and Optimization

- Performance and concurrency testing.
- Reservation and payment reconciliation.
- Monitoring, alerts, backups, and recovery testing.
- Analytics instrumentation and KPI dashboards.
- Usability testing and booking-flow optimization.

Release Gate: The MVP must pass critical reservation concurrency tests, payment reconciliation tests, security checks, and end-to-end booking tests before production launch.

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14. Risks and Mitigation

Risk	Impact	Mitigation
Outdated venue availability	Failed bookings and user frustration	Synchronize schedules and revalidate availability before reservation
Concurrent booking attempts	Double booking	Atomic slot holds and database-level consistency controls
Delayed payment notifications	Incorrect reservation status	Idempotent webhook processing and reconciliation
Unpaid Split Bill shares	Booking cancellation or financial loss	Clear deadlines, reminders, and captain settlement rules
Payment gateway downtime	Checkout abandonment	Clear status handling and supported retry options
Incorrect venue data	Reduced trust and conversion	Venue verification and controlled listing updates
Peak-hour traffic	Slow searches and failed bookings	Load testing, monitoring, and scalable infrastructure

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15. Open Questions and Product Decisions

The following decisions must be resolved before development is finalized:

1. **Booking confirmation:** Is the court confirmed only after full payment, or can the captain guarantee unpaid Split Bill shares?
2. **Split Bill deadline:** When must team members complete payment relative to the match start time?
3. **Slot-lock timing:** Does the five-minute hold begin when a slot is selected or when checkout begins?
4. **Late payment handling:** What happens if a payment succeeds after the reservation hold expires and the slot has been reassigned?

5. **Cancellation and refunds:** What refund rules apply to user cancellations, venue cancellations, and technical failures?
6. **Venue operations:** How will court owners create, update, block, and manage their schedules?
7. **Reviews:** Are reviews restricted to users with completed bookings?
8. **Payment methods:** Is credit card payment part of the MVP, or will the first release support only QRIS and bank virtual accounts?
9. **Notifications:** Which channels are required for booking confirmations and Split Bill reminders: email, in-app notifications, WhatsApp, or SMS?
10. **Performance targets:** What concurrent-user volume and response-time percentile will define successful peak-hour performance?

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16. Definition of Done

The Module 4 MVP is considered ready for release when:

- Users can register, authenticate, search for courts, and view availability.
- Users can reserve an available one-hour slot and complete checkout.
- Simultaneous reservation attempts cannot produce duplicate confirmed bookings.
- Expired slot holds are released correctly.
- Payment status is verified and synchronized with booking status.
- Split Bill allocations and outstanding balances are calculated accurately.
- Confirmed bookings generate valid digital tickets and QR codes.
- Booking history reflects the correct reservation and payment statuses.
- Critical error scenarios have been tested.
- Performance, security, and backup requirements have been validated.
- KPI tracking is implemented for the primary success metrics.
- All release-blocking defects are resolved.

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17. Final Product Summary

The Futsal Court Booking Platform aims to replace fragmented manual reservation processes with a centralized digital booking experience.

Its MVP focuses on four core capabilities:

1. **Discover:** Find and compare futsal courts using transparent venue information and search filters.
2. **Reserve:** View real-time availability and secure a slot through temporary reservation locking.
3. **Pay:** Complete digital payments and coordinate team expenses through Split Bill.
4. **Confirm:** Receive a digital booking ticket, QR code, and calendar invitation.

The primary measure of success is whether the platform enables team captains like Dane to secure reliable futsal reservations quickly, coordinate payments efficiently, and confidently return for future bookings.

Product Principle: Make every court booking transparent, secure, and predictable—from the first search to venue check-in.